HEIQ XMEMBRANE PROJECT

Model closure and material handoff

Frozen energy basis · direct-ERV gradient closure · AC null test · next experimental gate

Internal technical decision note · 31 August 2026

This is the current authoritative interpretation of the model study. It supersedes earlier reports where a positive passive surface-gradient direction was treated as a feasibility requirement or water diffusivity was presented as an independently sufficient material criterion. Earlier reports remain the chronological record.

Phase decision. The broad steady-state model phase is complete. Start the joint-property material search and controlled-flow coupon work now. Additional unfocused parameter sweeps are unlikely to change that decision. The project has identified conditional opportunity and the measurements needed to test it; it has not validated a membrane, selected a product architecture or demonstrated a net system energy saving.

Primary comparator	Passive ERV + remaining AC = 543.741 kW; cooling is included
Main material result	Performance is governed by a coupled property bundle, not by D or transport number alone
Direct-ERV result	Active uphill transport is thermodynamically admissible; film losses and the counter-current pinch determine the local gradient regime
Symmetric-current result	No useful quasi-static mean EO transport beyond the passive $N(0)$ state in the present model
Current/area boundary	No current-density acceptance ceiling and no exploratory membrane-area acceptance gate

Abstract

The frozen O80 equal-service basis is 1,166.400 kW for AC alone and 543.741 kW for passive ERV plus remaining AC. The post-ERV O30 single-common-voltage coordinate gives a 381.327-kW known subtotal and a 162.414-kW conditional window, while the full-duty direct-ERV coordinate gives 107.856 kW and a 435.884-kW (80.16%) conditional window. These differences exclude one or more of receiver/sink work, EO heat rejection, electrodes, manifolds, leakage, pressure drop and durability; they are not net savings.

The final direct-ERV audit resolves the gradient question. Positive passive surface direction is a regime diagnostic, not a thermodynamic gate for active EO. At the frozen O30 mechanism coordinate, the bulk activity margin is favourable, but two finite gas films reverse the surface margin to -0.00797. A 1.137-fold common increase in film capacity, or the equivalent diagnostic area increase at fixed duty, restores a non-negative surface direction. At full duty the bulk counter-current pinch is exactly zero, so no finite film or area multiplier produces a positive surface margin at finite flux; EO must act uphill at the pinch. All signed states close the applicable thermodynamic audit.

A matched quasi-static $+J/0/-J$ replay also closes the proposed symmetric-AC question within the current constant-drag model. O30 produces only 0.00152 kg/h of even-in-current transport above the passive state, equal to 0.348 ppm of the $+J$ water rate, while full duty gives zero. Thus a charge-balanced symmetric current does not provide useful mean EO pumping here; transient or rectified AC would require measured non-linearity, asymmetry or memory not present in this model. The correct next step is same-sample measurement of sorption, chemical water permeance, drag, wet conductivity, compressed-stack ASR, thickness, film exchange and coupled water flux in a controlled coupon. Candidate ranking and full-system claims remain open.

Closure statement. We know enough to stop the broad modelling loop. The remaining uncertainties are empirical and subsystem-specific; they should be resolved with materials, coupons and a receiver/thermal ledger rather than by searching a larger hypothetical parameter space.

1 Frozen energy and comparison basis

USE EQUAL SERVICE AND STATE EVERY OMITTED LOAD

System coordinate	Power [kW]	Difference vs ERV+AC [kW / %]	Claim class
AC only, O80	1,166.400	-622.659 / -114.51	Calibrated legacy comparator
Passive ERV + AC	543.741	0 / 0	Primary comparator: 34.359-kW ERV + 509.381-kW cooling
Post-ERV EO to O30, common 1.00 V	381.327	$+162.414$ / $+29.87$	Conditional incomplete ledger
Post-ERV EO to O30, uniform-current cross-check	379.516	$+164.225$ / $+30.20$	Conditional topology cross-check
Direct EO inside ERV, full humidity duty, 0.25 V	107.856	$+435.884$ / $+80.16$	Conditional architecture screen

For the post-ERV common-voltage row, 316.910 kW is remaining cooling and 64.417 kW is the ERV/active-device subtotal. For the direct-ERV row, 44.438 kW is remaining cooling and 63.419 kW is the known active-core/fan subtotal. The conservative post-ERV all-in break-even is 0.1567008 kWh/kg water removed (156.7 Wh/kg), not the historical 180-Wh/L shorthand.

The positive numbers in the difference column are **available conditional windows**, not measured or net system savings. The direct-ERV 80.16% result is the largest upside and the least complete system boundary. It assumes prescribed 0.75 sensible effectiveness, external EO-heat rejection, idealised hydraulics and an unmeasured inverse-design material vector.

The final common-voltage replay below 2 V separates numerical status from physical interpretation. On the refined 16-segment grid, O30 has 14 converged cases, 14 physical no-solution coordinates and 21 numerical/local unresolved coordinates; all seven sampled diffusivities nevertheless have a confirmed selected O30 case at 1.00–1.25 V. Full remaining duty has zero confirmed cases, 21 physical no-solution coordinates and 28 numerical/local unresolved coordinates. Across both segment counts, all 98 cases that encountered the legacy high-bracket flag were retried after increasing the configured numerical current bound from 50,000 to 5,000,000 A/m^2; none recovered. This rules out the configured numerical bound as the active limiter in those retries, but it does not convert the unresolved rows into a global physical no-go proof.

The post-ERV O30 known-core coordinate removes 1,228.27 kg/h at 30.057 kW (40.864 kg/kWh known-core FOM). The direct-ERV full-duty coordinate transfers 4,827.15 kg/h with a 63.419-kW known active-core/fan subtotal. These FOMs have different device boundaries and must not be used as product efficiencies without the omitted loads.

2 Direct EO in the ERV: corrected gradient interpretation

ACTIVE FEASIBILITY AND PASSIVE ASSISTANCE ARE DIFFERENT QUESTIONS

For net molar water flux $N > 0$ from source to receiver, the finite-film model obeys the segment identity

$$\Delta a_{\mathrm{surface}} = \Delta a_{\mathrm{bulk}} - N\left(\frac{1}{K_s} + \frac{1}{K_r}\right), \tag{1}$$

where K_s and K_r are the source- and receiver-side film capacities. The numerical identity error is below 3.0×10^{-15} in the closure replay. A negative $\Delta a_{\mathrm{surface}}$ says that EO is transporting uphill locally; it is not by itself a violation of thermodynamics. The active state is admissible when the electrical/chemical coupling and dissipation audit closes.

Coordinate	Bulk Δa	Surface Δa	Area	V_t	Interpretation
Direct-ERV O30 mechanism point	$+0.0581$ to $+0.0969$	-0.00797 to -0.00536	24,604 m^2	0.025 V	Passive bulk assistance exists, but film losses reverse the local membrane-surface direction
Direct-ERV full duty	0 at the pinch	-0.0822 to -0.0495	34,425 m^2	0.25 V	No passive bulk headroom at the pinch; finite-flux transfer is locally uphill

At O30, multiplying both film capacities by 1.137 restores $\min(\Delta a_{\mathrm{surface}}) = 0$. The equivalent area diagnostic at fixed duty is 27,974 m^2. Recovering the former $+0.02$ robustness margin requires 1.733 times the film capacity or 42,641 m^2. These are diagnostic crossover values, not area targets. At full duty, increasing film capacity or area only approaches zero from below; no finite multiplier creates a positive surface margin because the bulk margin is zero.

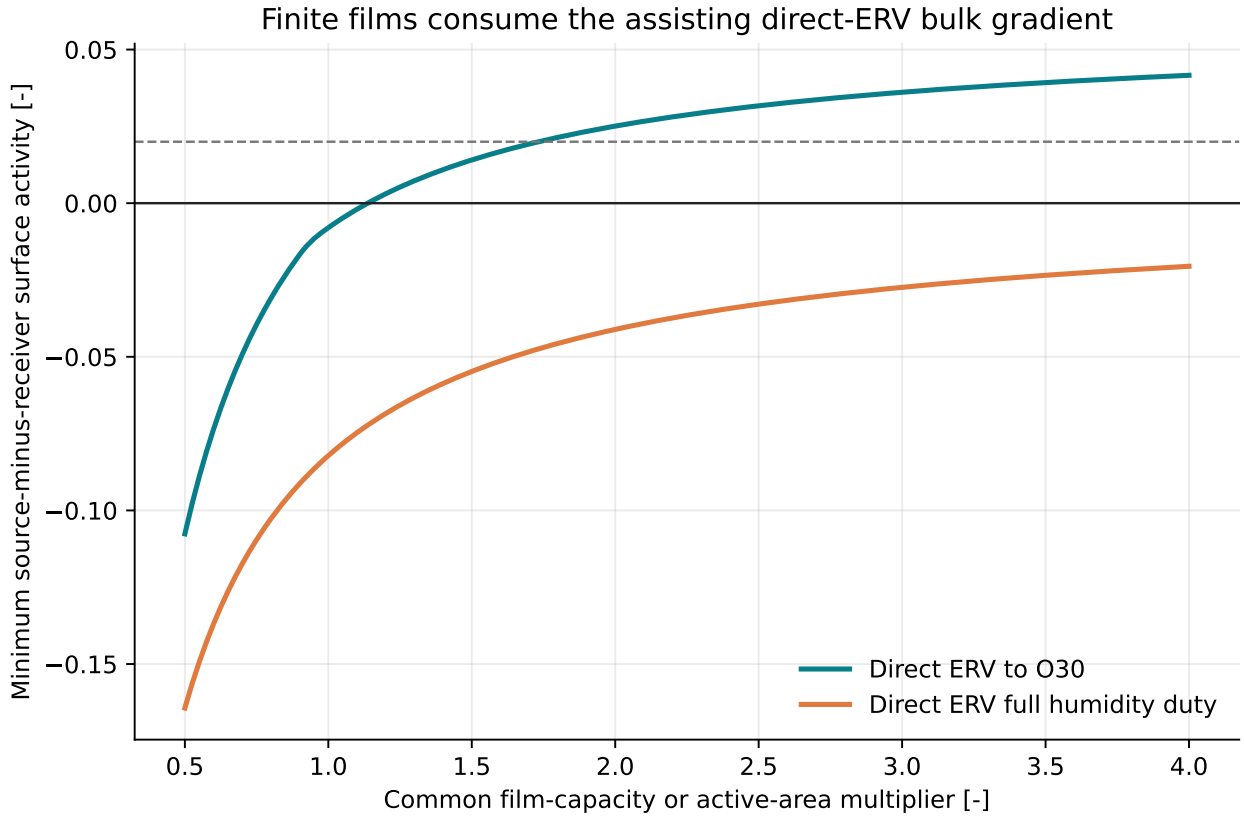


Figure 1: Direct-ERV film/area gradient envelope. The O30 mechanism point has a finite surface-direction crossover; the full-duty counter-current pinch has none at finite local flux. Area is diagnostic, not an acceptance gate.

Water-polarisation work accounts for up to 99.877% of the 0.025-V O30 terminal voltage and 99.990% of the 0.25-V full-duty voltage. This is why "low resistance" alone cannot identify a good membrane: the chemical water work, passive permeance and interface conditions must be evaluated together.

Resolved gradient statement. Direct EO in the ERV can exploit a favourable bulk gradient where one exists and can actively cross an adverse surface gradient when electrical work closes the dissipation balance. The potential benefit is therefore real at model level, but the film coefficients and pinch location are first-order experimental inputs.

3 Symmetric-current AC: quasi-static closure

A GRADIENT CAN BREAK SYMMETRY, BUT THIS MODEL CONTAINS NO USEFUL RECTIFIER

The final test evaluates the same local constitutive law at $+J$, zero current and $-J$, then averages the two signed-current states. The current amplitudes are those of the frozen positive-DC coordinates. This is a quasi-static, current-symmetric replay; it is not a frequency-domain or electrode model.

Coordinate	$+J$ water [kg/h]	$N(0)$ [kg/h]	Cycle mean [kg/h]	Increment over $N(0)$ [kg/h]	Cycle electrical power [kW]
Direct-ERV O30	4,365.918	0.415151	0.416669	0.00151783	20.092
Direct-ERV full duty	4,827.147	0	0	0	17.944

The O30 even-in-current increment is only 3.48×10^{-7} of the $+J$ water rate (0.348 ppm); full duty is exactly zero in the present model. Consequently, a charge-balanced symmetric current adds no useful mean EO transport beyond passive $N(0)$ while still dissipating electrical power.

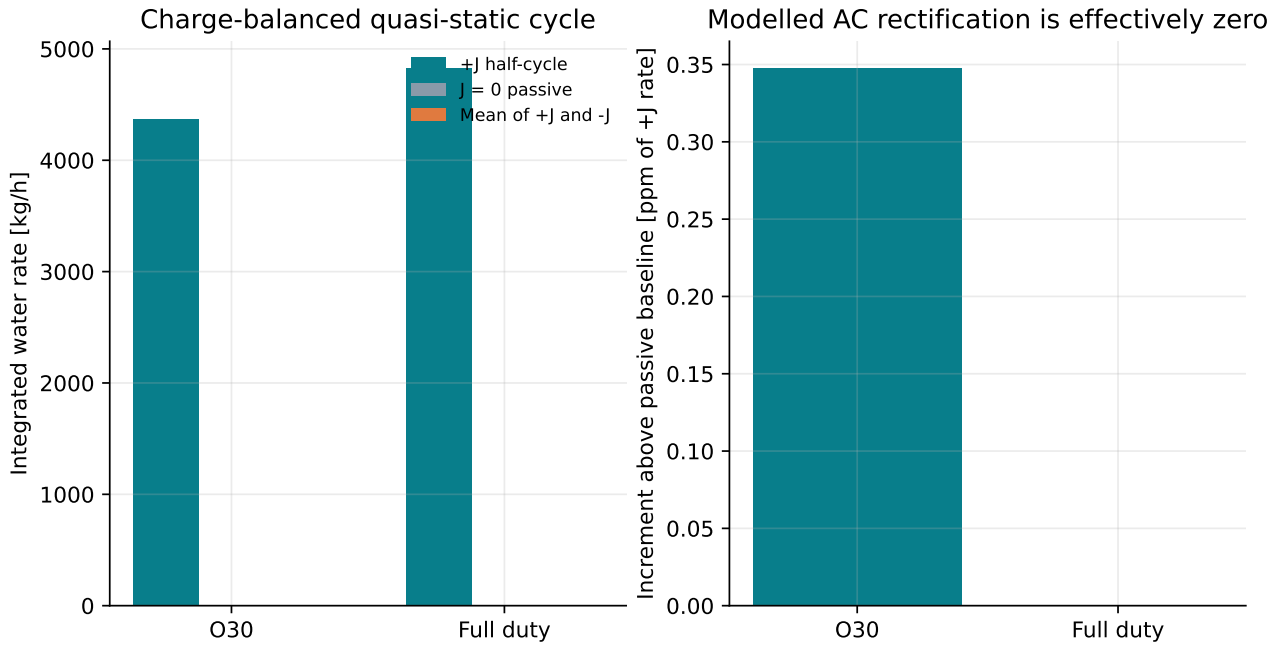


Figure 2: Matched $+J/0/-J$ quasi-static replay. Positive DC transports the required water, whereas signed-current averaging cancels the polarity-odd EO contribution and leaves essentially the passive state.

This result does not exclude AC transport based on time-dependent hydration, hysteresis, asymmetric interfaces, nonlinear electrode kinetics, induced charge or a deliberately graded membrane. It says those mechanisms are absent from the current evidence and should not trigger another modelling campaign before dynamic measurements exist. Also, zero-mean voltage need not imply zero-mean current when a chemical-potential offset is present.

4 What the material search should optimise

THE IDENTIFIABLE QUANTITY IS A COUPLED PROPERTY BUNDLE

The focused $D = 10^{-14}$–10^{-9} m^2/s sweep held water capacity, thickness, drag, conductivity, ASR and activity exponents fixed. It therefore proves a strong *one-factor sensitivity at that coordinate*; it does not prove that a dry-state scalar diffusivity is universally dominant. In a sorption-based membrane, the passive term contains the chemical water permeance, schematically

$$\mathcal{P}_{w,\mathrm{chem}} \propto \frac{D(a,T)}{L}\frac{\partial C_w}{\partial a_w} \quad \text{or an equivalent chemical-potential mobility.} \tag{2}$$

The derivative, thickness and interfaces are therefore inseparable from D when ranking candidates.

Same-sample property	Why it is required	Exploratory direction / method
Sorption $C_w(a,T)$, hysteresis and $\partial C_w/\partial \mu_w$	Converts humidity into chemical loading/work and closes the passive mobility	Full adsorption/desorption curves across roughly $a = 0.05$–0.9 and 23–32°C
$D(a,T)$ or Maxwell–Stefan mobility, with definition	Separates kinetics from capacity and quantifies back-/co-diffusion	The post-ERV frozen screen flags $D \lesssim 10^{-11}$ and prefers $\lesssim 10^{-12}$ m^2/s; this is not a universal gate
EO drag $n_d(a,T,j)$	Sets polarity-odd water transported per charge	High and reproducible; the inverse-design coordinate is about 100 mol H_2O/mol charge, not a minimum pass/fail value
Wet through-plane conductivity and compressed-stack ASR	Sets ohmic work and exposes contact losses	Same hydration/ion form as drag and diffusion; inverse-design coordinates are 23.17 S/m and $10^{-5}\ \Omega\,\mathrm{m}^2$
Dry/wet thickness, swelling, creep and strength	Couples permeance and resistance to durability	Thin but stable; inverse-design dry thickness is 36.9 μm
Source/receiver exchange coefficients	Determines whether bulk assistance survives at membrane surfaces	Measure each side versus humidity, flow and gap; direct-ERV O30 has a 1.137-fold crossover
Coupled net water flux and charge	Tests whether the property bundle coexists	Two-sided water balance under $+J/0/-J$ and repeated polarity blocks

Post-ERV polishing and direct-ERV integration need not prefer the same material. Post-ERV EO generally benefits from suppressing adverse chemical permeance. Direct ERV can use favourable passive transfer before the pinch, so indiscriminately minimising permeance can remove useful free transport. Every candidate should be replayed against both architecture boundaries after its same-sample data are available.

Do not construct a synthetic "best membrane" by combining D, drag, conductivity and thickness from different materials or conditioning states. Nafion is not the target, area is not yet capped, there is no 21-A/m^2 gate, and a transport number above 100 is not sufficient by itself.

5 Closed questions, open gates and handoff

STOP BROAD MODELLING; CONVERT UNCERTAINTY INTO MEASUREMENTS

Question or gate	Status	Decision
Energy comparator and cooling inclusion	CLOSED	Use 543.741 kW passive ERV+AC as the primary equal-service basis
Active transport with a negative surface margin	CLOSED	Thermodynamically admissible when the coupling/dissipation audit closes; passive direction remains a diagnostic
Direct-ERV film and pinch behaviour	CLOSED	O30 has a finite 1.137-fold crossover; full duty has none at finite flux because the bulk pinch is zero
Symmetric-current benefit in the present model	CLOSED	No useful mean EO increment; defer transient AC until measured dynamics justify it
Exploratory current density and area	CLOSED	Report solved values; apply neither an A/m^2 ceiling nor an area acceptance gate now
Start material search and coupon work	GO	Begin now using the joint-property and same-sample requirements
Rank a material candidate	OPEN	No candidate yet has the complete same-sample property set
Full post-ERV duty on one common bus below 2 V	OPEN	No confirmed sampled point: 21 physical no-solution and 28 numerical/local unresolved refined coordinates; test staging/zoning only after material data
Select post-ERV versus direct-ERV architecture	OPEN	Requires measured film, receiver, thermal, pressure-drop and durability evidence
Claim net system or liquid-product performance	NO-GO	Requires a complete measured receiver/heat/hardware ledger and long-term operation

5.1 Immediate experimental handoff

Use the 25-cm^2 controlled-flow coupon as the primary transport experiment. The current practical basis is a roughly 110×110-mm fixture, 2-mm flow gaps and 6 L/min per side, followed by 3/6/12-L/min and 1/2/4-mm gap repeats. Run zero-current controls and an ABBA polarity sequence; measure both source loss and receiver gain, charge and voltage, and keep the membrane differential pressure within ±5 Pa. Double seals, foil blanks and aperture/perimeter masks are required to separate active-area transport from bypass and edge leakage.

For each coupon, freeze candidate ID, batch, ion form, conditioning, temperature, humidity and wet thickness. Measure sorption, D/mobility, drag, wet conductivity, compressed ASR, film sensitivity, swelling/creep and the coupled two-sided water balance on that same material. A quiescent closed cell may be used only as a transient diagnostic; the present model does not support a fixed-boundary quiescent steady-state rate.

Stop criterion reached. No further broad model study is required before materials are sought. Continue modelling only as a replay of measured candidate data, a pre-registered design of the coupon experiment, or closure of a named subsystem such as the receiver/thermal ledger.

6 Claim boundaries and canonical evidence

WHAT THIS REPORT DOES AND DOES NOT ESTABLISH

All active results are steady-state and conditional. The direct-ERV model prescribes sensible effectiveness and external EO-heat rejection and does not resolve conjugate temperature/humidity feedback, condensation, manifold pressure drop, leakage, electrode kinetics, ageing or control dynamics. The signed-current replay is quasi-static; it does not predict frequency response, hysteresis, induced-charge EO, electrode rectification or transport in a spatially graded membrane. No full-coupled or commercial claim is enabled.

Permitted wording is: *the model identifies a conditional energy window and a measurable membrane-property region*. Do not state that 162.4 kW or 435.9 kW has been saved, that the 80.16% direct-ERV window is a net system saving, that full duty is impossible, or that a candidate membrane has been identified.

Canonical evidence paths

- `outputs/fom_assessment_v1/system_fom.csv` and `device_fom.csv`: frozen equal-service and known-core energy/FOM ledgers;

- `outputs/workshop_common_voltage_diffusivity_v1/`: post-ERV common-voltage diffusivity replay below 2 V;
- `outputs/direct_active_eo_erv_study_v1/study/`: direct active-ERV cases, thermodynamic/engineering screens and convergence;
- `outputs/direct_erv_gradient_closure_v1/`: film identity, area/film sweep, signed $+J/0/-J$ replay, figures, audit and source manifest;
- `outputs/project_closure_check_v1/`: project-level thermodynamic reclassification, common-voltage replay and decision gates;
- `outputs/eo_vapor_cell_sizing_v1/` and `outputs/eo_vapor_boundary_analysis_v1/`: coupon sizing, detectability and quiescent/transient boundary analysis;
- `reports/workshop_pack_2026-08-30/MATERIAL_SEARCH_BRIEF.md`: material-property schema, non-gates and supplier/-coupon screening sequence.

Every numerical output directory above contains an audit and/or source manifest. Audit PASS means that the declared numerical and identity checks passed; it does not mean that a material, module or technology passed validation.

End of model-closure report. This document freezes the current interpretation of the declared model evidence as of 31 August 2026.